ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Agricultural Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept other relevant, correct answers.

QUESTION 1

1.1 1.1.1 B ✓✓ 1.1.2 B ✓✓ 1.1.3 A ✓✓ 1.1.4 C ✓✓ 1.1.5 C ✓✓ (10)

1.2 1.2.1 Alloy ✓✓ 1.2.2 Straining post / end post ✓✓ 1.2.3 Borehole ✓✓ 1.2.4 Welding ✓✓ 1.2.5 Electric fence ✓✓ (10)

[20]

QUESTION 2

2.1 Any FIVE: wear protective clothing (overalls, safety boots, goggles) ✓; keep the workshop clean and tidy ✓; use tools only for their intended purpose ✓; switch off machines before adjusting them ✓; know where the fire extinguisher and first-aid kit are ✓; no loose clothing or jewellery near machines; never work alone with dangerous machines; ventilate the workshop (5)

2.2 Ferrous metals contain iron ✓, e.g. steel or cast iron ✓. Non-ferrous metals contain no iron ✓, e.g. copper, aluminium or zinc ✓. (4)

2.3 Iron in steel reacts with oxygen and water (moisture) ✓✓ to form rust (iron oxide). Prevention, any TWO: paint ✓; galvanising (zinc coating) ✓; oil or grease; keep dry; use stainless steel (4)

2.4 Cutting, any TWO: hacksaw ✓; tin snips ✓; bolt cutter; angle grinder (power tool). Measuring, any ONE: steel rule / tape measure / vernier caliper / square ✓ (3)

2.5 Any FOUR: clean tools after use ✓; oil metal parts to prevent rust ✓; store tools in a dry, organised place ✓; sharpen cutting tools ✓; replace or repair damaged handles; check tools before use (4)

[20]

QUESTION 3

3.1 Any THREE: keep livestock in and predators or stock thieves out ✓; divide the farm into camps for rotational grazing ✓; protect crops from animals ✓; mark boundaries; keep animals off roads (3)

3.2.1 $2 \times (400 + 250)$ ✓ $= 1\,300$ m ✓ (2)

3.2.2 $1\,300 \times 5$ ✓ $= 6\,500$ m ✓ (2)

3.2.3 $1\,300 \div 5$ ✓ $= 260$ droppers ✓ (2)

3.2.4 $6\,500 \div 500 = 13$ rolls ✓; $13 \times$ R1 250 ✓ $=$ R16 250 ✓ (3)

3.3 Any logical steps: mark the position ✓; dig a deep hole ✓; place the post upright (check with a spirit level) and set it in concrete or ram the soil firmly ✓; fit a stay (brace) to resist the pull of the wires ✓. (4)

3.4 Any FOUR: bricks / blocks ✓; cement and concrete ✓; steel / corrugated iron ✓; wood (poles, timber) ✓; stone; thatch (4)

[20]

QUESTION 4

4.1 Any FOUR, 1 mark for the source and 1 for the type: solar ✓ renewable ✓; wind ✓ renewable ✓; diesel / petrol ✓ non-renewable ✓; Eskom electricity from coal ✓ non-renewable ✓; biogas (renewable); hydropower (renewable); wood (renewable) (8)

4.2.1 $750\ \text{W} = 0{,}75\ \text{kW}$ ✓; $0{,}75 \times 6$ ✓ $= 4{,}5\ \text{kWh}$ ✓ (3)

4.2.2 $4{,}5 \times 30 = 135\ \text{kWh}$ ✓; $135 \times \text{R}2{,}50$ ✓ $= \text{R}337{,}50$ ✓ (3)

4.3 Any THREE: free energy after installation ✓; works during load shedding / in remote areas without the grid ✓; renewable and does not pollute ✓; low maintenance (3)

4.4 Any ONE rule, explained, e.g.: switch off the power at the main switch before working on a circuit ✓✓, because electricity can cause serious burns or death ✓. (Others: use qualified electricians; keep electrical equipment dry; do not overload plugs; check cables for damage.) (3)

[20]

QUESTION 5

5.1 Any FOUR: boreholes ✓; rivers and streams ✓; dams ✓; rainwater harvested in tanks ✓; springs; municipal supply (4)

5.2 Any TWO: windmill (wind pump) ✓; electric submersible pump ✓; solar pump; hand pump; centrifugal pump; diesel pump (2)

5.3 $\text{r} = 2 \div 2 = 1\ \text{m}$ ✓; $\text{V} = \pi\text{r}^2\text{h}$ ✓ $= 3{,}14 \times 1^2 \times 1{,}5$ ✓ $= 4{,}71\ \text{m}^3$ ✓ $= 4\ 710\ \ell$ ✓ (5)

5.4 Any FOUR: drip irrigation ✓; irrigate early in the morning or evening ✓; fix leaking pipes and troughs ✓; collect rainwater ✓; use mulch; float valves in troughs; recycle water (4)

5.5 Water can contain harmful bacteria or parasites ✓ that make animals sick ✓; it can contain too much salt, nitrates or other chemicals ✓ that reduce growth and production ✓; regular testing finds problems early so they can be treated ✓. (5)

[20]

TOTAL: 100

END OF MARKING GUIDELINE